Class 18

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Background

Pertussis (a.k.a. whooping cough) is a common lung infection caused by bacteria B. Pertussis.

This can infect all ages but is most sever for those under 1 year of age.

The CDC track the number of reported cases in US:We can "scrape" this data with the **datapasta** package

```
data <- data.frame(
      stringsAsFactors = FALSE,
           check.names = FALSE,
          `cdc.<-.data.frame(` = c("year = c(","1922L,","1923L,","1924L,","1925L,",
                                   "1926L,","1927L,","1928L,","1929L,",
                                   "1930L,","1931L,","1932L,","1933L,","1934L,",
                                   "1935L,","1936L,","1937L,","1938L,",
                                   "1939L,","1940L,","1941L,","1942L,","1943L,",
                                   "1944L,","1945L,","1946L,","1947L,",
                                   "1948L,","1949L,","1950L,","1951L,","1952L,",
                                   "1953L,","1954L,","1955L,","1956L,",
                                   "1957L,","1958L,","1959L,","1960L,","1961L,",
                                   "1962L,","1963L,","1964L,","1965L,",
                                   "1966L,","1967L,","1968L,","1969L,","1970L,",
                                   "1971L,","1972L,","1973L,","1974L,",
                                   "1975L,","1976L,","1977L,","1978L,","1979L,",
                                   "1980L,","1981L,","1982L,","1983L,",
                                   "1984L,","1985L,","1986L,","1987L,","1988L,",
                                   "1989L,","1990L,","1991L,","1992L,",
                                   "1993L,","1994L,","1995L,","1996L,","1997L,",
                                   "1998L,","1999L,","2000L,","2001L,",
                                   "2002L,","2003L,","2004L,","2005L,","2006L,",
                                   "2007L,","2008L,","2009L,","2010L,",
                                   "2011L,","2012L,","2013L,","2014L,","2015L,",
```

```
                                        "2016L,","2017L,","2018L,","2019L,",
                                        "2020L,","2021L,","2022L,","2023L,","2024L,",
                                        "2025L,"),","cases = c(","107473,",
                                        "164191,","165418,","152003,","202210,",
                                        "181411,","161799,","197371,","166914,",
                                        "172559,","215343,","179135,","265269,",
                                        "180518,","147237,","214652,","227319,",
                                        "103188,","183866,","222202,","191383,","191890,",
                                        "109873,","133792,","109860,","156517,",
                                        "74715,","69479,","120718,","68687,",
                                        "45030,","37129,","60886,","62786,","31732,",
                                        "28295,","32148,","40005,","14809,",
                                        "11468,","17749,","17135,","13005,","6799,",
                                        "7717,","9718,","4810,","3285,","4249,",
                                        "3036,","3287,","1759,","2402,","1738,",
                                        "1010,","2177,","2063,","1623,","1730,",
                                        "1248,","1895,","2463,","2276,","3589,",
                                        "4195,","2823,","3450,","4157,","4570,",
                                        "2719,","4083,","6586,","4617,","5137,",
                                        "7796,","6564,","7405,","7298,","7867,",
                                        "7580,","9771,","11647,","25827,",
                                        "25616,","15632,","10454,","13278,","16858,",
                                        "27550,","18719,","48277,","28639,",
                                        "32971,","20762,","17972,","18975,","15609,",
                                        "18617,","6124,","2116,","3044,","7063,",
                                        "22538,","21996,")","")")
        )
```

```
cdc <- tibble::tribble(
  ~year, ~cases,
  1922, 107473,
  1923, 164191,
  1924, 165418,
  1925, 152003,
  1926, 202210,
  1927, 181411,
  1928, 161799,
  1929, 197371,
  1930, 166914,
  1931, 172559,
  1932, 215343,
  1933, 179135,
```

```
1934, 265269,
1935, 180518,
1936, 147237,
1937, 214652,
1938, 227319,
1939, 103188,
1940, 183866,
1941, 222202,
1942, 191383,
1943, 191890,
1944, 109873,
1945, 133792,
1946, 109860,
1947, 156517,
1948, 74715,
1949, 69479,
1950, 120718,
1951, 68687,
1952, 45030,
1953, 37129,
1954, 60886,
1955, 62786,
1956, 31732,
1957, 28295,
1958, 32148,
1959, 40005,
1960, 14809,
1961, 11468,
1962, 17749,
1963, 17135,
1964, 13005,
1965, 6799,
1966, 7717,
1967, 9718,
1968, 4810,
1969, 3285,
1970, 4249,
1971, 3036,
1972, 3287,
1973, 1759,
1974, 2402,
1975, 1738,
```

```
1976, 1010,
1977, 2177,
1978, 2063,
1979, 1623,
1980, 1730,
1981, 1248,
1982, 1895,
1983, 2463,
1984, 2276,
1985, 3589,
1986, 4195,
1987, 2823,
1988, 3450,
1989, 4157,
1990, 4570,
1991, 2719,
1992, 4083,
1993, 6586,
1994, 4617,
1995, 5137,
1996, 7796,
1997, 6564,
1998, 7405,
1999, 7298,
2000, 7867,
2001, 7580,
2002, 9771,
2003, 11647,
2004, 25827,
2005, 25616,
2006, 15632,
2007, 10454,
2008, 13278,
2009, 16858,
2010, 27550,
2011, 18719,
2012, 48277,
2013, 28639,
2014, 32971,
2015, 20762,
2016, 17972,
2017, 18975,
```

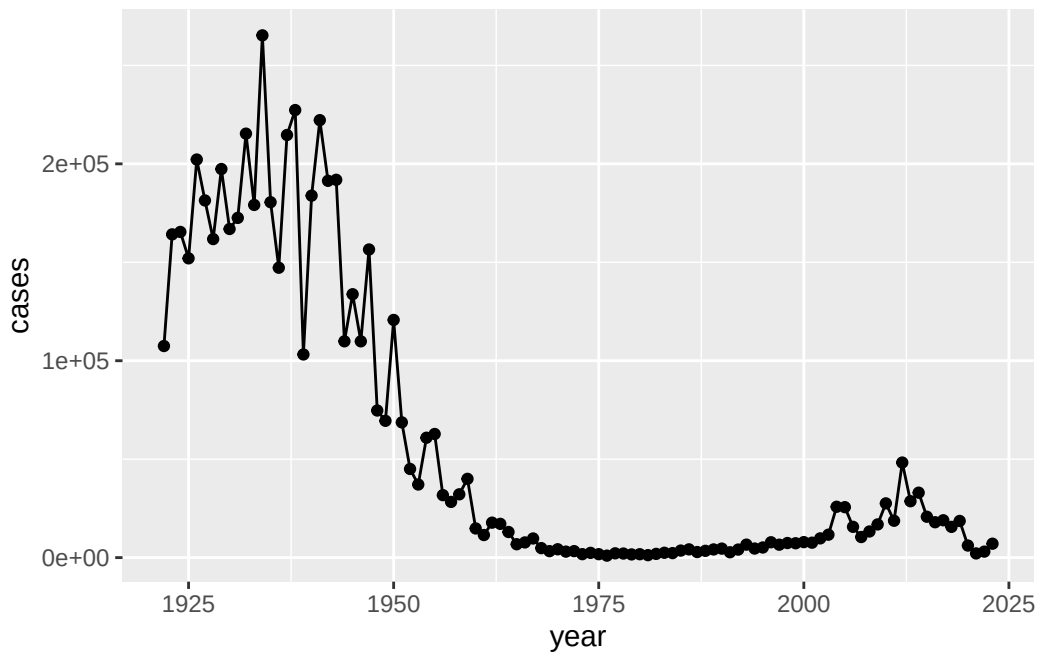
```
  2018, 15609,
  2019, 18617,
  2020, 6124,
  2021, 2116,
  2022, 3044,
  2023, 7063
)
```

Q1. Make a plot of `year` vs `cases`

```
library(ggplot2)
```

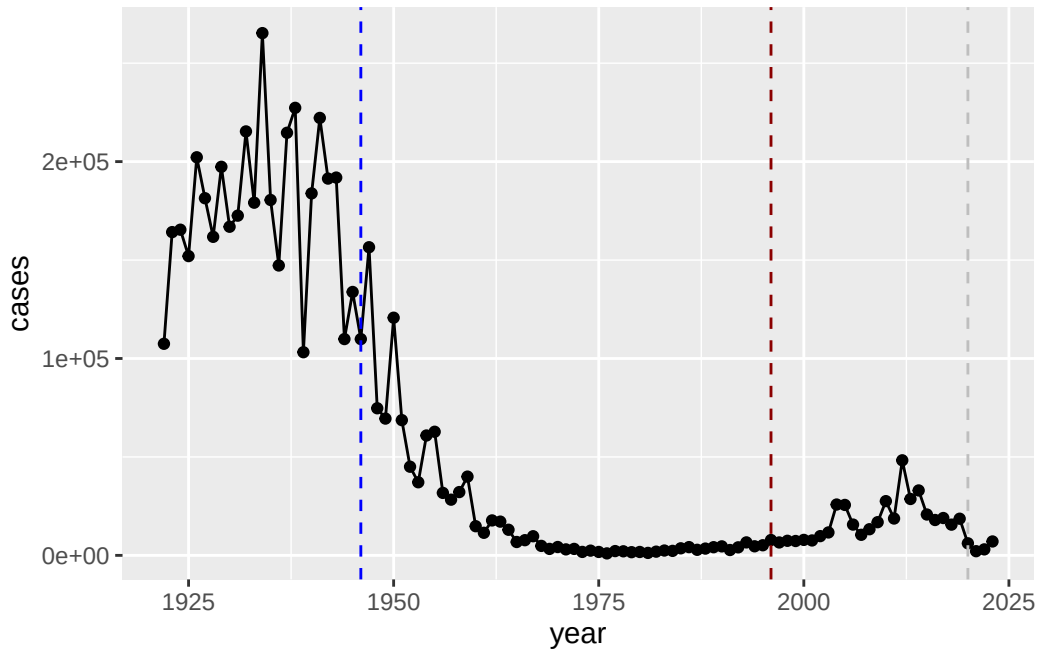
```
library(ggplot2)

ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line()
```



Q2. Add some major milestones including the first wP vaccine roll-out (1946), the switch to the newer aP vaccine (1946), the switch ti the newer aP vaccine (1996), the COVID years (2020)

```r
ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 1946, col= "blue", lty=2) +
  geom_vline(xintercept = 1996, col="darkred", lty=2) +
  geom_vline(xintercept = 2020, col="grey", lty=2)
```



Answer: There were high case numbers in the pre 1940s; then there was a dramatic drop from of cases that time to the 1960-1970s. At this time it it contunited to be very small (or much smaller) amount of cases than before 1946 /1940s.

> Q3. Describe what happened after happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend?

In the short term, aP was able to combat Pertussis, however the vaccine was not good at doing this for a long term effect. Which is most likely what happened and is shown in the data. The immunity from aP decreases much faster (protection decreases).

Why is this vaccine-preventable disease on the upswing? To answer this question we need to investigate the mechanisms underlying waning protection against pertussis. This requires evaluation of pertussis-specific immune responses over time in wP and aP vaccinated individuals.

CMI-PB project

Computational Models of Immunity - Pertussis Boost project aims to provide the scientific community with this very information.

They make their data available via JSON format reutrning API. We can read this in Twith the `read_json` function from the **jsonlite** package:

```
library(jsonlite)

subject <- read_json("https://www.cmi-pb.org/api/subject", simplifyVector = TRUE)

head(subject)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          2          wP         Female Not Hispanic or Latino White
3          3          wP         Female                Unknown White
4          4          wP           Male Not Hispanic or Latino Asian
5          5          wP           Male Not Hispanic or Latino Asian
6          6          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset
1    1986-01-01    2016-09-12 2020_dataset
2    1968-01-01    2019-01-28 2020_dataset
3    1983-01-01    2016-10-10 2020_dataset
4    1988-01-01    2016-08-29 2020_dataset
5    1991-01-01    2016-08-29 2020_dataset
6    1988-01-01    2016-10-10 2020_dataset
```

Q4. How many aP and wP infancy vaccinated subjects are in the dataset?

```
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q5. How many Male and Female subjects/patients are in the dataset?

```
table(subject$biological_sex)
```

```
Female   Male
   112     60
```

WP is higher for on go back!!!!

> Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc..)? (In terms of race and gender is this dataset representative of the US population?)

```r
table(subject$race, subject$biological_sex)
```

```
                                          Female Male
  American Indian/Alaska Native                0    1
  Asian                                       32   12
  Black or African American                    2    3
  More Than One Race                          15    4
  Native Hawaiian or Other Pacific Islander    1    1
  Unknown or Not Reported                     14    7
  White                                       48   32
```

Answer: There is 32 Asian females and 12 Asian males, 48 White females and 32 White males, 2 Black females and 3 Black males. Then there is 0 American Indian/Alaska Native females and 1 American Indian/Alaska Native male. There is one female and one male Native Hawaiian or Other Pacific Islander. 15 female and 4 male individuals of More Than One Race. There are 14 females and 7 males that so not have there race reported.

Let's read the next two:

```r
specimen <- read_json("http://cmi-pb.org/api/v5_1/specimen", simplifyVector = TRUE)

ab_titer <- read_json("http://cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = TRUE)
```

```r
head(specimen)
```

```
  specimen_id subject_id actual_day_relative_to_boost
1           1          1                           -3
2           2          1                            1
3           3          1                            3
4           4          1                            7
```

```
5           5          1                           11
6           6          1                           32
  planned_day_relative_to_boost specimen_type visit
1                             0         Blood     1
2                             1         Blood     2
3                             3         Blood     3
4                             7         Blood     4
5                            14         Blood     5
6                            30         Blood     6
```

```r
head(ab_titer)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection
1 UG/ML                 2.096133
2 IU/ML                29.170000
3 IU/ML                 0.530000
4 IU/ML                 6.205949
5 IU/ML                 4.679535
6 IU/ML                 2.816431
```

To analyze this data we need to first "join" (merge/link) the different tables so we have all the data in one place not spread across different tables.

We can use the `*_join()` family of functions from **dplyr** to do this

```r
library(dplyr)
```

```
Attaching package: 'dplyr'


The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
meta <- inner_join(subject, specimen)
```

```
Joining with `by = join_by(subject_id)`
```

```r
head(meta)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          1          wP         Female Not Hispanic or Latino White
3          1          wP         Female Not Hispanic or Latino White
4          1          wP         Female Not Hispanic or Latino White
5          1          wP         Female Not Hispanic or Latino White
6          1          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset specimen_id
1    1986-01-01    2016-09-12 2020_dataset           1
2    1986-01-01    2016-09-12 2020_dataset           2
3    1986-01-01    2016-09-12 2020_dataset           3
4    1986-01-01    2016-09-12 2020_dataset           4
5    1986-01-01    2016-09-12 2020_dataset           5
6    1986-01-01    2016-09-12 2020_dataset           6
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                            1                             1         Blood
3                            3                             3         Blood
4                            7                             7         Blood
5                           11                            14         Blood
6                           32                            30         Blood
  visit
1     1
2     2
3     3
4     4
5     5
6     6
```

```r
abdata <- inner_join(ab_titer, meta)
```

```
Joining with `by = join_by(specimen_id)`
```

```r
head(abdata)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 UG/ML                 2.096133          1          wP         Female
2 IU/ML                29.170000          1          wP         Female
3 IU/ML                 0.530000          1          wP         Female
4 IU/ML                 6.205949          1          wP         Female
5 IU/ML                 4.679535          1          wP         Female
6 IU/ML                 2.816431          1          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
5 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
6 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
5                           -3                             0         Blood
6                           -3                             0         Blood
  visit
1     1
2     1
3     1
4     1
5     1
6     1
```

> Q. What Antibody isotypes are measured for these patients? Q11. How many specimens (i.e. entries in abdata) do we have for each isotype?

```
table(abdata$isotype)
```

```
  IgE   IgG  IgG1  IgG2  IgG3  IgG4 
 6698  7265 11993 12000 12000 12000 
```

Answer: IgE: 6,698 specimens IgG: 7,265 specimens IgG1: 11,993 specimens IgG2: 12,000 specimens IgG3: 12,000 specimens IgG4: 12,000 specimens

> Q. what antigens are reported? Q12. What are the different $dataset values in abdata and what do you notice about the number of rows for the most "recent" dataset?

```
table(abdata$antigen)
```

```
    ACT   BETV1      DT   FELD1     FHA  FIM2/3   LOLP1     LOS Measles     OVA 
   1970    1970    6318    1970    6712    6318    1970    1970    1970    6318 
    PD1     PRN      PT     PTM   Total      TT 
   1970    6712    6712    1970     788    6318 
```

```
table(abdata$dataset)
```

```
2020_dataset 2021_dataset 2022_dataset 2023_dataset 
       31520         8085         7301        15050 
```

Answer: There is 2020_dataset: 31520
2021_dataset: 8085
2022_dataset: 7301
2023_dataset: 15050.

In recent years there is much less rows than in earlier years (2020 has much more than 2023 and 2022). This may show that earlier years had much more data collected.

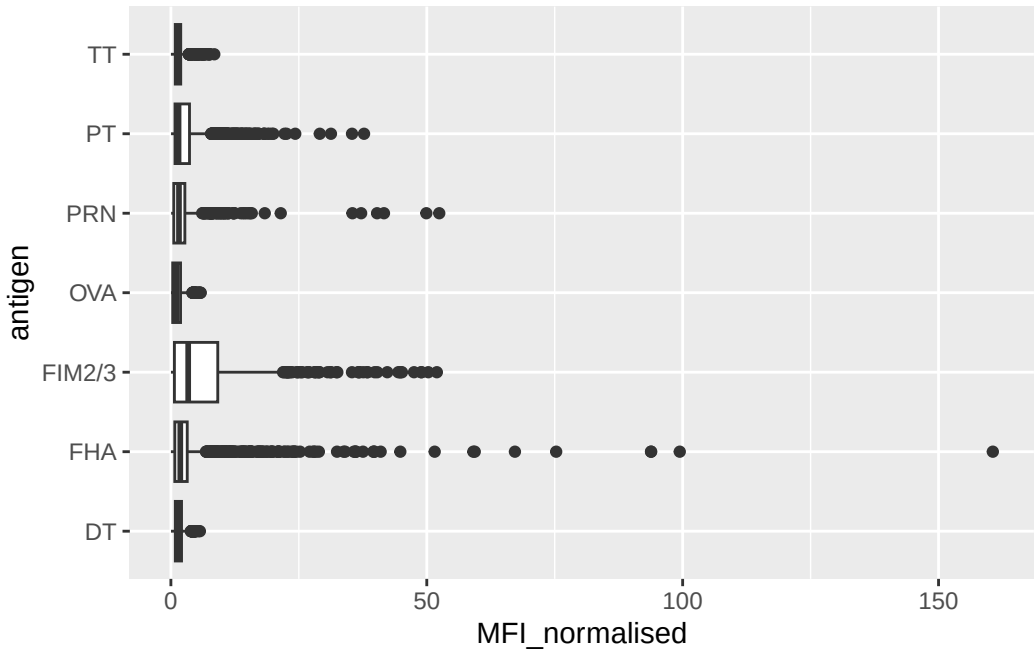
Let's focus on the IgG antigen and make a plot of MFI_normalized for all antigens.

```r
igg <- abdata |>
  filter(isotype == "IgG")

head(igg)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgG                TRUE      PT   68.56614       3.736992
2           1     IgG                TRUE     PRN  332.12718       2.602350
3           1     IgG                TRUE     FHA 1887.12263      34.050956
4          19     IgG                TRUE      PT   20.11607       1.096366
5          19     IgG                TRUE     PRN  976.67419       7.652635
6          19     IgG                TRUE     FHA   60.76626       1.096457
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 IU/ML                 0.530000          1          wP         Female
2 IU/ML                 6.205949          1          wP         Female
3 IU/ML                 4.679535          1          wP         Female
4 IU/ML                 0.530000          3          wP         Female
5 IU/ML                 6.205949          3          wP         Female
6 IU/ML                 4.679535          3          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4                Unknown White    1983-01-01    2016-10-10 2020_dataset
5                Unknown White    1983-01-01    2016-10-10 2020_dataset
6                Unknown White    1983-01-01    2016-10-10 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
5                           -3                             0         Blood
6                           -3                             0         Blood
  visit
1     1
2     1
3     1
4     1
5     1
6     1
```

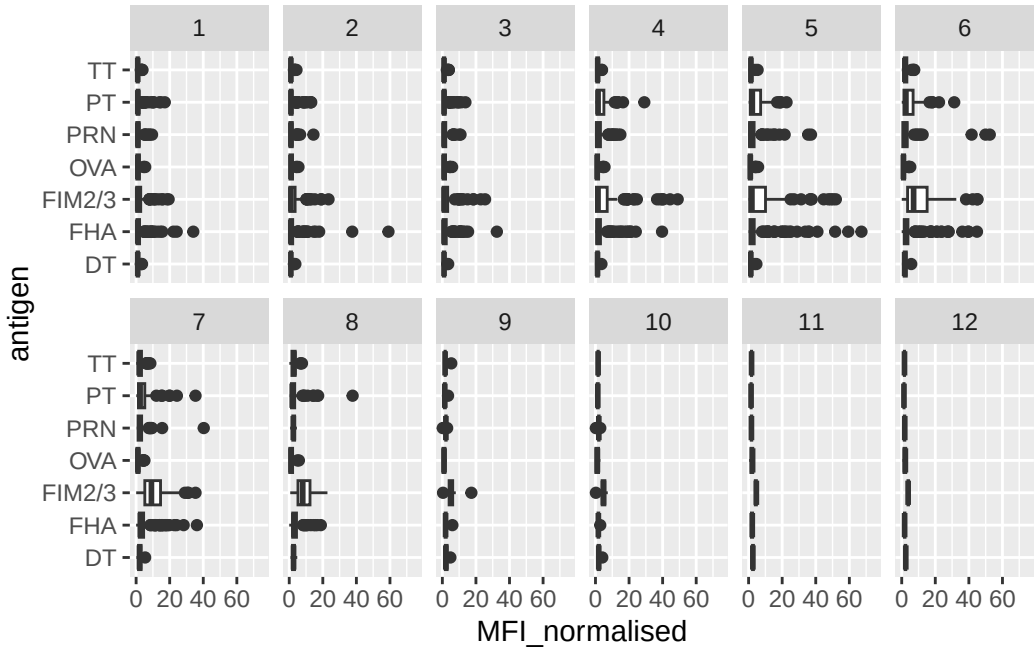
```r
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot()
```



Q 13.

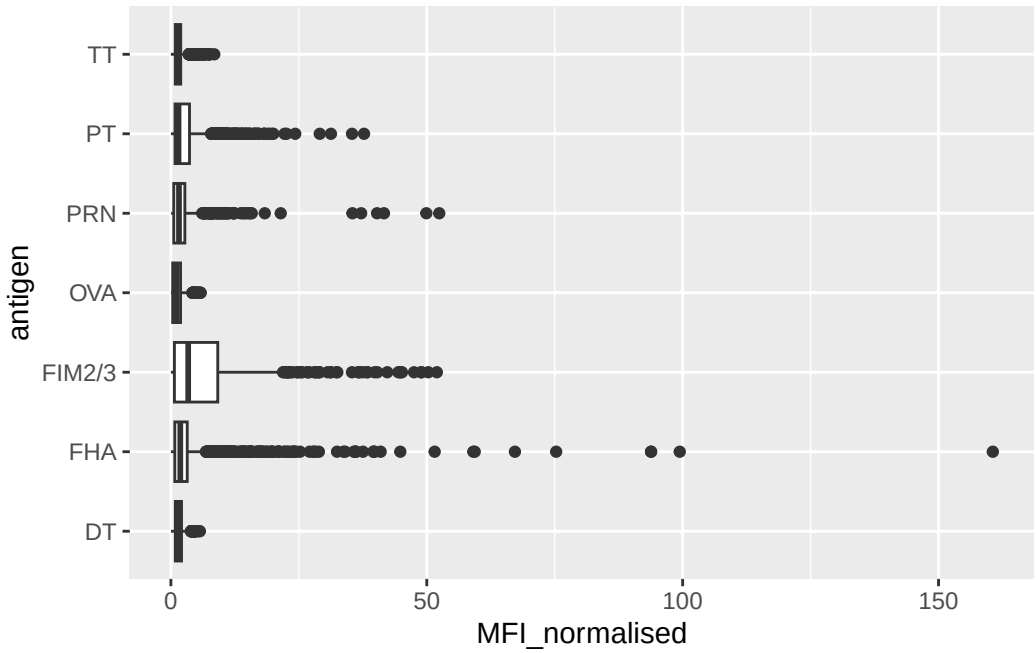
```r
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot() +
  xlim(0,75) +
  facet_wrap(vars(visit), nrow=2)
```

```
Warning: Removed 5 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



Q. Is there a differenece for aP vs wP indivduals with these values?

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot()
```

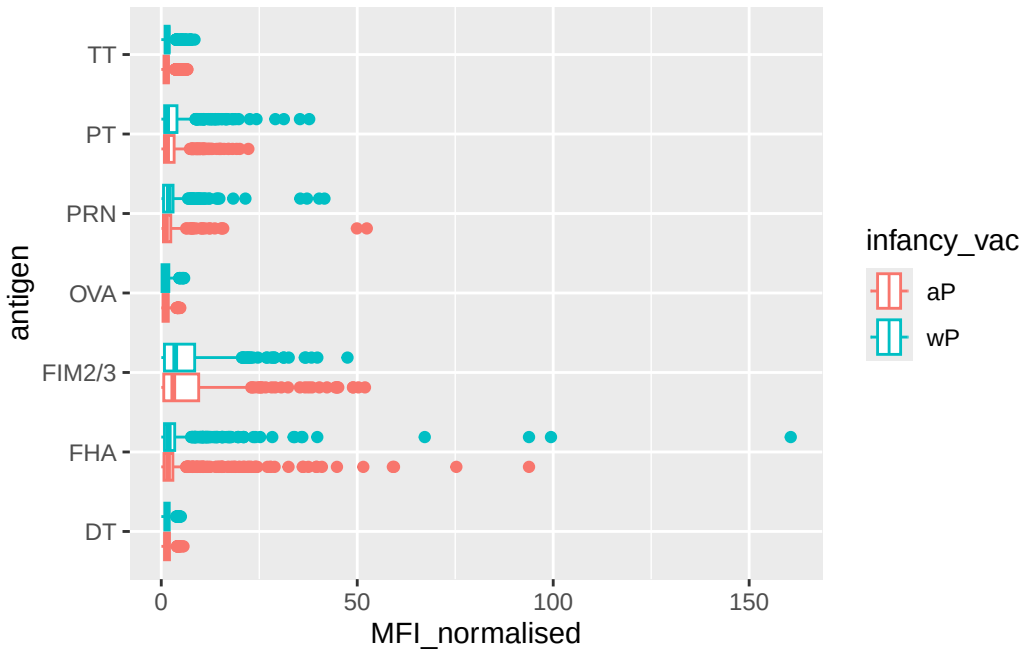



```
facet_wrap(~infancy_vac)
```

```
<ggproto object: Class FacetWrap, Facet, gg>
    attach_axes: function
    attach_strips: function
    compute_layout: function
    draw_back: function
    draw_front: function
    draw_labels: function
    draw_panel_content: function
    draw_panels: function
    finish_data: function
    format_strip_labels: function
    init_gtable: function
    init_scales: function
    map_data: function
    params: list
    set_panel_size: function
    setup_data: function
    setup_panel_params: function
    setup_params: function
    shrink: TRUE
```

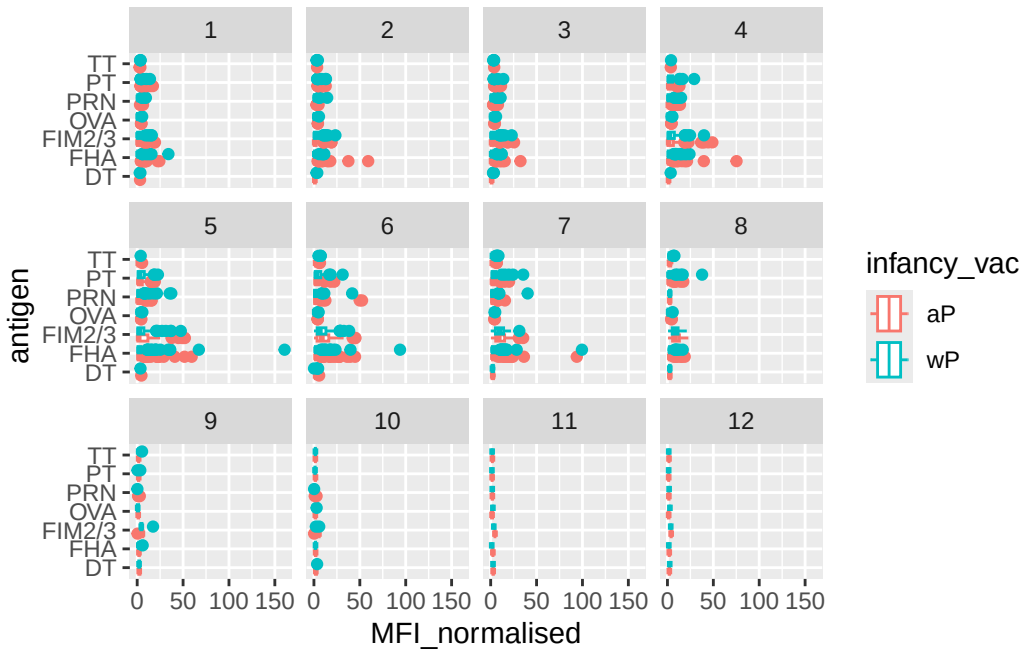
```
    train_scales: function
    vars: function
    super:  <ggproto object: Class FacetWrap, Facet, gg>
```

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac) +
  geom_boxplot()
```



Q. Is there a temprol response - i.e do valus increase our decrease over time?

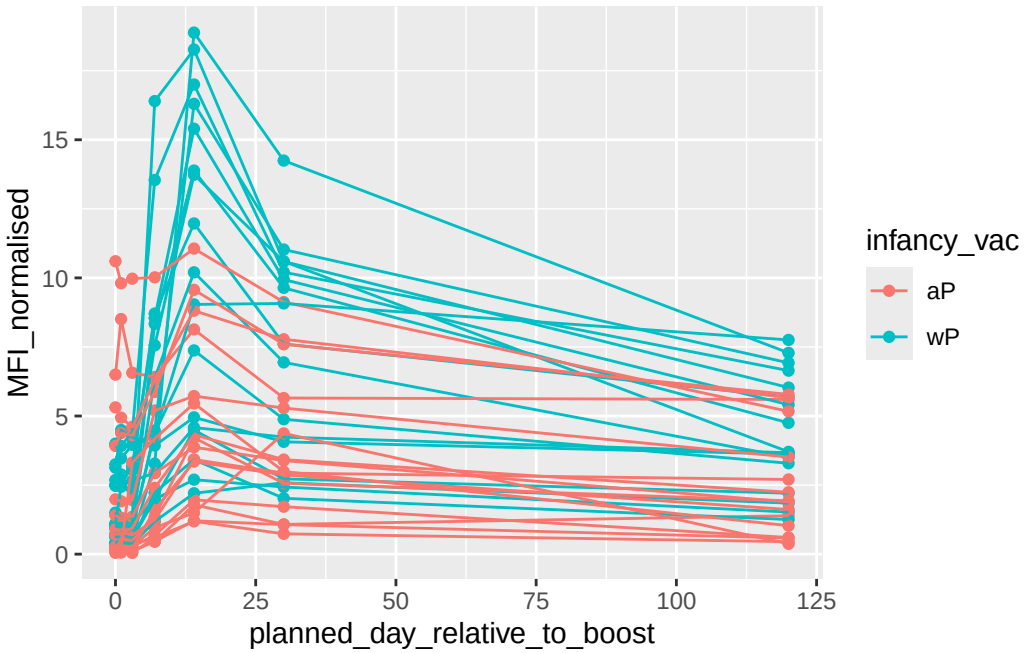
```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac) +
geom_boxplot() +
  facet_wrap(~visit)
```



Focus on "PT" pertussis Toxin antigen

```
pt.igg.21 <- igg |> filter(antigen == "PT",
                           dataset== "2021_dataset")
```

```
ggplot(pt.igg.21) +
  aes(planned_day_relative_to_boost, MFI_normalised,
      col=infancy_vac,
      group= subject_id) +
  geom_point() +
  geom_line()
```

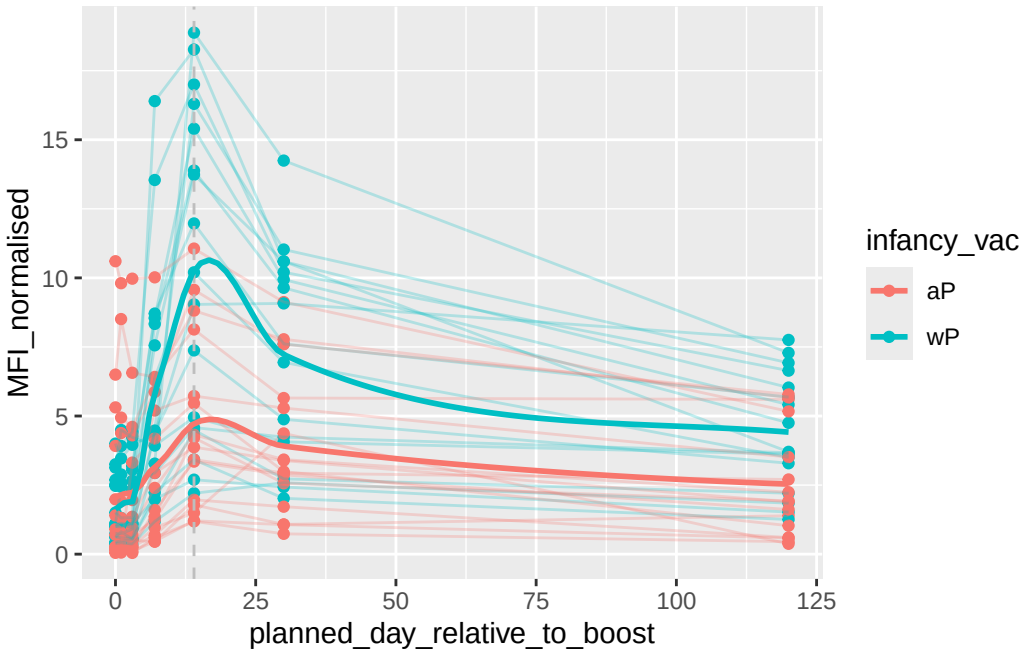



```
geom_vline(xintercept = 14, col="grey", lty=2)
```

```
mapping: xintercept = ~xintercept
geom_vline: na.rm = FALSE
stat_identity: na.rm = FALSE
position_identity
```

```
ggplot(pt.igg.21) +
  aes(planned_day_relative_to_boost, MFI_normalised,
      col = infancy_vac,
      group = subject_id) +
  geom_point() +
  geom_line(alpha=0.25) +
  geom_vline(xintercept = 14, col = "grey", lty = 2) +
  geom_smooth(aes(col=infancy_vac, group=NULL),
    method = "loess",
    se = FALSE, span=0.4
  )
```

```
`geom_smooth()` using formula = 'y ~ x'
```


MFI_normalised
15
10
5
0
0
25
50
75
100
125
planned_day_relative_to_boost
infancy_vac
aP
wP